

ANISH KHURANA

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2021 – Apr 2026

EXPERIENCE

University of Waterloo – Undergraduate Research Assistant

Jan 2026 – Apr 2026

Supervisor: Dr. Gennaro Notomista

- Designed and implemented a ROS2-based multi-robot orchestration platform for DJI RoboMaster robots using YAML fleet configurations and live UDP network discovery.
- Built automated launch/session management workflows enabling parameterized deployment and lifecycle control of multiple robots.
- Diagnosed Docker networking and DDS middleware failures impacting distributed ROS2 communication systems.

Zoocasa – QA Automation Developer

Sep 2025 – Dec 2025

- Built and maintained automated unit and integration test suites using **Jest** and **React Testing Library**, **increasing test coverage by ~40%** and reducing manual regression effort.
- Executed end-to-end testing for major releases, identifying **50+** defects across staging and production environments.
- Contributed directly to production features across **React/Next.js** frontend and **Ruby on Rails/Go** backend services to improve data integrity within a **multi-tenant architecture**.
- Built a hackathon prototype featuring an end-to-end AI pipeline that classified room types from real estate listing photos, generated 360° panoramas using Stable Diffusion, and delivered interactive property tours through a web-based viewer.

Microart Services Inc – Software Engineering Intern

Sep 2023 – Dec 2023 | Jan 2025 – Apr 2025

- Built guided in-app firmware update workflows for Bluetooth-enabled devices, enabling **100% of supported users** to complete updates successfully.
- Improved PostgreSQL/Supabase RPC performance by 33%** within scalable backend services handling high-frequency production data retrieval.
- Optimized and expanded user review submission flows through improved navigation logic and database interactions, **reducing completion time by ~30%**.
- Developed a high-volume JavaScript data pipeline, **scraping and processing 100K+ records** into PostgreSQL systems.

VCT Group – Software Engineering Associate

May 2024 – Aug 2024

- Refactored a major production website section using TypeScript, React, and Java, **reducing technical debt by ~30%**.
- Modernized frontend/backend integrations and applied **domain-driven design** principles to improve maintainability and streamline full-stack workflows.

Okos Smart Homes - Software Development Intern

Jan 2023 - Apr 2023

- Delivered forgot-password functionality for web and mobile platforms using secure authentication best practices.
- Developed and maintained internal tools used by **20+ project managers** and **200+ technicians**.

PROJECTS

AccessWise - Final Year Capstone Design Project (github.com/programmer812/AccessWise)

- Built a full-stack accessibility mapping platform leveraging **computer vision** to analyze public imagery and detect urban features, enabling scalable evaluation of campus accessibility.
- Designed ML-powered pipelines combining public datasets and crowdsourced feedback to support scalable AODA-aligned accessibility evaluation.

Power Code (devpost.com/software/power-code)

- Built a real-time multiplayer coding duel platform using React, FastAPI, Material UI, and Convex for **Hack the North**.

Math Notes to Latex Converter (github.com/rohanvedula/PaZaR)

- Developed **Raspberry Pi + OpenCV + OCR** pipeline converting handwritten math notes into LaTeX-generated PDFs.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, Go, C/C++, SQL, C#, HTML/CSS

Frameworks/Systems: React, Next.js, React Native, Node.js, FastAPI, Flask, Rails, GraphQL, Docker, Git, ROS2

Databases/Cloud: PostgreSQL, MongoDB, Firebase, Supabase, AWS, GCP

Systems/Testing: Linux/Unix, Jest, Playwright, Pytest, React Testing Library, TensorFlow, PyTorch, Scikit-learn